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Exploring Evolution Using Museum Collections

INTRODUCTION

Museum collections are often underappreciated as warehouses for preserving and storing specimens that have been collected in the field, with many collected in the late 1800s and early 1900s. However, museums serve a vital role in ongoing evolutionary and ecological research that allows scientists to verify the identity of species, discover new species, and preserve specimens for later observation. Furthermore, these collections are critical for tracing historical trends in organism size, shape, coloration, life histories and more.

In this lab, you will work as a research scientist to identify insect specimens captured in the field, verify your findings by comparing these unknown specimens to tangible and digital museum specimens, and investigate the relatedness of your unknown specimens by mapping them in a phylogenetic tree.

PART ONE: USING A DICHOTOMOUS KEY TO IDENTIFY UNKNOWN ORGANISMS

When new specimens are collected in the field, one of the first things curators and scientists must do is identify and classify the organism. Classification is a way of separating a large group of closely related organisms into smaller subgroups. For example, dogs are subdivided into many different breeds because of their distinguishing characteristics. A dichotomous key is used to classify organisms, using a list of characteristics to identify the organism. In this investigation, you will use a key to identify different adult species of insects.

A **dichotomous key** is a written tool used to determine the taxonomic identification of plants, animals, soils, etc. and is based on a two (*di*-) characteristic pattern, or couplet. These characteristics are generally opposites of one another and satisfy an either/or logic system. For example, an object may be round or not round (square, rectangular, octagonal), it may be metal (aluminum, steel, copper) or not metal (wood, plastic, bone). The first characteristic in the key is usually the most broad and encompassing characteristic as to split the unknown group into two large groups. As one progresses through the key, each descriptor couplet becomes more specific, for example the hairs on a leaf are long versus short. Ultimately the precise identification to the unknown organisms *Genus*, or *Genus species*, is accomplished. Most field guides utilize this format, enabling amateur birders to identify what species they just saw or budding mycologists to identify a mushroom they have picked to determine whether or not it is edible.

How to use a key?

Each descriptive couplet in a dichotomous key has both a number and a letter. The numbers are sequential integers (1, 2, 3, 4) with each successive number being associated with a different descriptive phrase (ie. is/is not metal). The letters are just **a** and **b**; the two options of the descriptive phrase (a = metal, b= not metal). If the object or organism being examined does not belong in the ***a*** category, it should belong in the ***b*** one. Note that identification keys are generally based only on visual descriptive characteristics - morphology. They do not necessarily show phylogenetic, evolutionary relationships. Two organisms may look quite similar and could be grouped together in a field identification key, but when examined in greater detail using biochemical or genetic information they may be found to be unrelated to the extreme.

To further explain the use of a dichotomous key, imagine that you have been given a box containing these objects: large nail, thumb tack, safety pin, paper clip, wood pencil, crayon, bic pen, cotton swab. Here is your key:

1a. composed of only metal ……………................... go to 2
1b. composed of materials other than metal ………. go to 5
2a. one end has a sharp pointed tip …………………….. go to 3
2b. end is not sharp or pointed …………………………… ***paperclip***
3a. opposite end is flat ……………………………………… go to 4
3b. opposite end is not flat ………………..……………… ***safety pin***
4a. is longer than 2 centimeters ………………........... ***nail***
4b. is shorter than 2 centimeters …………..……………***thumb tack***
5a. is a writing tool ……………………………………………. go to 6
5b. is not a writing tool ……………………………………... ***cotton swab***
6a. mark is erasable …………………………………………… ***pencil***
6b. mark is not erasable …………………………………… go to 7
7a. mark is waxy ………………………………………………. ***crayon***
7b. mark is not waxy ………………………………………….. ***pen***

Reading couplet 1, you divide your items into two piles: (nail, thumb tack, safety pin, paper clip) and (pencil, crayon, pen, cotton swab). To identify the metal items, go to couplet 2. The non-metal items will be identified at number 5 a bit later in these instructions. Three of your items have a pointed tip (nail, thumb tack, safety pin), one does not – the paper clip. So at 2b you have identified the first object as a paper clip. Of your sharp (2a) objects, two have an opposing end that is flat – which objects are they? One is not flat, so at 3b you have identified the safety pin. Couplet 4 describes the length of the objects that have a pointed end and a flat end. The nail is longer than 2 centimeters (4a) and the thumb tack is shorter than 2 centimeters (4b). You have successfully identified all of your metal objects!

To identify the remaining objects, the not-metal ones, read couplet 5. Note that you have skipped 2, 3 and 4 for these items because those couplets are associated only with the metal objects. Three of your objects are writing tools (5a) (pencil, crayon, pen). The one that is not a writing tool (5b) is identified as a cotton swab. Couplet 6 differentiates between written marks that are erasable (6a) and not erasable (6b). Only one item makes an erasable mark and it is thusly identified to be a pencil. The final two items are examined with regard to the final couple 7. One item makes a waxy mark (7a) and is a crayon, the mark made by the other is not waxy so it is a pen. You have now identified your non-metal items as well!

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Why use a key?

Sometimes you can identify organisms by comparing them to pictures in field guides or similar nature apps. While field guides are an important tool, the use of a key is essential for accurate identification of an organism. Sometimes organisms, even from different Orders, can look exactly alike (convergent evolution? Perhaps). For example, many flies (order Diptera) look almost like wasps (order Hymenoptera). Using your key, you will find that a fly has 1 pair of wings, whereas wasps have 2 pairs of wings.

In this part of the lab, student groups will use a key to identify unknown insects (preserved in plastic blocks) to *Order.* More detailed identifications to *Family, Genus,* and *species* are beyond the scope of this course, but can be accomplished using more appropriate guides. A brief insect dichotomous key can be found in *Appendix A* at the end of this assignment. Because insects are so small, differentiating among species, families and even orders is often difficult. However, examination beneath a hand lens or dissecting microscope may help you to see many of the characteristics mentioned in the key.

Procedure

1. First, number your specimens so they can be easily referenced throughout this procedure.
2. Look closely at your unknown insects and make a list of similarities and differences in characteristics that you can observe, such as texture of wings, number of legs, spines on legs and body shape for each of your insects. **Write down you descriptions in Table 1 below.**
3. Review the dichotomous key in *Appendix A* so that you become familiar with the structure of the key. Use Figure 1 and Figure 2 below to familiarize yourself with the numerous descriptive terms associated with insect anatomy.
4. Starting with Insect 1, follow the pathway or numbers along the dichotomous key that correspond to the characteristics of Insect 1. For example, if Insect 1 has wings (1a), then go to 5. Read all the characteristics for 5a and 5b until you find the correct characteristic for Insect 1. If the Insect 1 does not have wings (1b), then go to 2. Read all the characteristics for 2a and 2b, until you find the correct characteristic for Insect 1.
5. Continue following the pathway or numbers until an order name for Insect 1 is determined.
6. Repeat steps 2 – 4 until all the insects have been identified. **Write the *Order* for each insect in Table 1 below.**
7. Verify your insect identifications with your GSI before proceeding to Parts 2 and 3.

Table 1: Unknown insect characteristics and identification

Insect	Characteristics	Order
Insect 1		
Insect 2		
Insect 3		
Insect 4		
Insect 5		

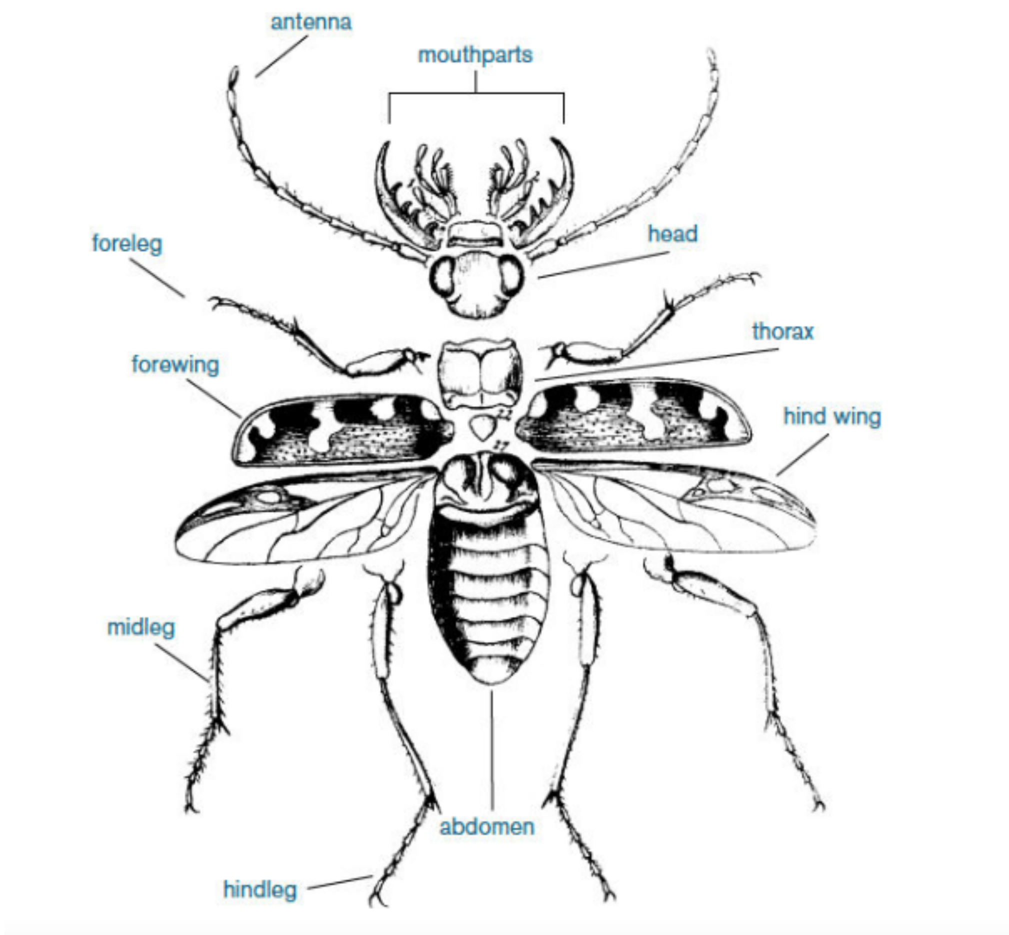

Figure 1: Body parts of an insect. (Source: Nicolls 2020)

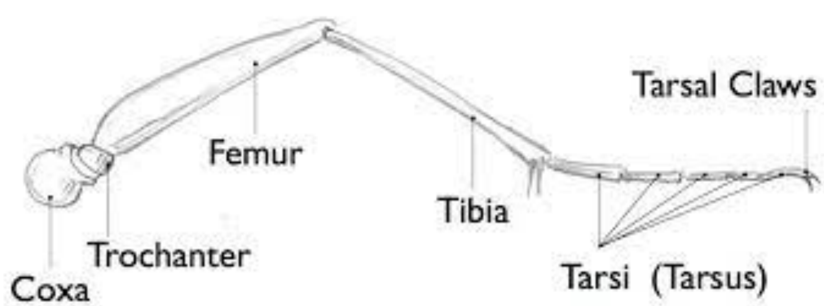

Figure 2: Parts of an insect leg. (Source: John Muir Laws 2017)

PART 2: COMPARE SPECIMENS TO MUSEUM SPECIMENS

Once identified, field specimens are compared to other preserved specimens in the collection. It is important to verify your identification from PART 1 and note physical and morphological similarities and differences. Sometimes further morphological and genetic comparisons reveal that a new species has been discovered!

View the preserved collection specimens along the side bench. For each of your specimens, find 2-3 similar preserved specimens in the collection (preferable from same Order).

QUESTIONS

1. Compare and contrast morphological differences between your specimens and those in the collection (i.e., size, shape, color, structures, leg or antennae length, etc.). You are encouraged to use illustrations in your description.

Specimen 1:

Specimen 2:

Specimen 3:

Specimen 4:

2. Besides age and sex, what factors might influence the morphological differences observed among individuals from the same species in a museum collection?

PART 3: BUILDING AN EVOLUTIONARY TREE

Now that you have verified the classification of your collected specimens you are ready to begin an evolutionary analysis of how these species may be related. You will use a phylogenetic tree generator, phyloT, to develop an evolutionary tree of your specimens. phyloT uses taxonomic and genetic data from the Genome Taxonomy Database and the NIH National Center for Biotechnology Information (NCBI) database to generate a tree in the users selected format. From this generated tree, you will investigate the evolutionary relationships among your different insect orders, and establish common ancestors.

As you may recall, evolutionary trees are hypotheses about the evolutionary relationships among organisms based on shared characteristics. So far, you have used physical and morphological features to characterize your individual specimens. Before you use the tree generator, you will develop a hypothesis (a tree) about the relatedness of your insect orders based on morphological data you collect from Parts 1 and 2. Examine all of your specimens closely. Compare and contrast their morphological similarities and differences, and then draw how you predict them to be related in an evolutionary tree. Be sure to consider maximum parsimony (i.e., K.I.S.S.) in the design of your tree. You may also wish to use a character table (as shown in lecture) to help form your hypothesis.

Draw your evolutionary tree hypothesis below. Then, write a brief paragraph explaining what characteristics you used to develop your tree.

Now that you have formulated a hypothesis using physical characteristics of your specimens, you will use the *phyloT* tree generator to compare your insect orders using genomic data.

Procedure:

1. Go to https://phylot.biobyte.de/
   (Note: You do not need to register with phyloT to use the phylogenetic tree generator.)
2. Scroll down to the NCBI tree elements area and in the "Search NCBI taxonomy" box, type the order name of your first specimen. The search engine should autopopulate a list of taxonomic IDs. Be sure to select the one that refers to your insects ORDER. For example, if you type in "Diptera" you will select the *Diperta (order, ID:7147).*
3. Repeat step 2 for all of your insects until all of the NCBI taxonomic IDs are listed in the box.
4. Under *Tree options*, make sure the following parameters are selected:
   a. Node identifiers = scientific names
   b. Internal nodes = expanded
   c. polytomy = yes
   d. Interrupt at = no interruption
   e. Format = newick
   f. Ignore errors = no
5. Name your tree in the *Tree or file name* box.
6. Click the ***Visualize in iTOL*** button.
7. Once the evolutionary tree is generated, go to the Control Panel and to the *Advanced* tab. Under "Branch metadata display," click **Display** for Node IDs. This will add the most common ancestor for each Branch node in the tree.
8. Use the Zoom-in and Zoom-out buttons to the left of your tree to investigate the nodes and reveal common ancestry among your insect orders.

Draw your evolutionary tree below. You do not need to include all of the Basal taxon (base of tree) information but should include at least 3 branch points and common ancestry.

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Questions:

1. How does your initial hypothesis above compare to the generated evolutionary tree?

2. What is the most recent common ancestor for ALL of your insect orders?

3. Are any of your insect orders sister taxa?

4. Are any of your insects an outgroup?

5. Some lineages branched many times and are represented by many living species. Discuss the ecological conditions that you think might result in the rapid diversification of some lineages (A real world example would be the rapid diversification of the mammals at the beginning of the Cenozoic, right after the dinosaurs went extinct.)

6. Some lineages changed very little over time. A good example of this would be "living fossils" like the horseshoe crab or cockroach. Again, discuss the ecological conditions that might result in this sort of long-term evolutionary stasis.

7. Some species went extinct without leaving descendants. In the real world, what factors might increase or decrease the probability of a species going extinct?

8. Some beetles and flies have antler-like structures on their heads, much like male deer. The existence of antlers in beetle, fly, and deer species with strong male-male competition is an example of **convergent evolution.** Give two other examples of convergent evolution between insects and other organisms.

REFERENCES

Foulk, G. (2004) Student handout ~ Dichotomous key for insects<<http://virtual.clemson.edu/groups/SCLife/lesson%20plans/adult%20insects/student%20handout%20_terr.insects_.pdf> > Clemson Pp. 1-11.

Kaskel, A., Hummer, P., Kennedy, J. and R. Oram (1994). Investigations living systems Laboratory Manual. Glencoe. McGraw-Hill. Columbus, Ohio. Pp. 133-136.

Nicolls M. 2020. *Similar But Different In the Animal Kingdom*. Strategic Book Publishing and Rights Co. ISBN: 978-1-68181-941-9

Walsh LL, Giffen CJ, Thompson CW. 2019. Teaching Biodiversity with Museum Specimens in an Inquiry-Based Lab. CourseSource. https://doi.org/10.24918/cs.2019.45

APPENDIX A: KEY TO COMMON INSECT ORDERS

Note: there are over 30 insect orders, some of which you will never encounter. The following key includes only 24 of the more common orders.

1a. Without wings; all of the abdominal segments visible in a top view of the insect 2
1b. With wings; wings may be difficult to see because the flying wings are hidden by hard wing covers. In these cases, the wing covers lie over the back and hide all of or parts of the abdomen 17

2a. Without legs, eyes, or antennae; living under a waxy or cottony covering and occurring in colonies firmly attached to tree twigs, fruit, or leaves ....HEMIPTERA
2b. Legs, antennae, and (usually) eyes present 3
2c. Has 8 legs and is not an insect at all!...Class: Arachnida Order: Araneae

3a. Abdomen ending in three long, thread-like tails; antennae long .... THYSANURA
3b. Abdomen without long tails; antennae may be long or short 4

4a. Antennae are shorter than the head, and not easily seen; body flattened from side-to-side or from top-tobottom; parasites on animals 5
4b. Antennae longer than the head, easily seen; not usually parasites 7

5a. Body flattened from side-to-side; legs long, able to jump; with sucking mouthparts.... SIPHONAPTERA
5b. Body flattened from top-to-bottom; legs short and not able to jump 6

6a. Abdomen sac-like and without distinct segments; eyes clearly visible; tarsi 5-segmented; about 1 cm long; sheep parasites .... DIPTERA
6b. Abdominal segments distinct; eyes small or absent; tarsi 1- to 2-segmented; less than 1/8 inch long ....PHTHIRAPTERA

7a. Body strongly constricted between the thorax and abdomen ....HYMENOPTERA
7b. Thorax and abdomen broadly joined 8

8a. Body scaly; coiled tongue sometimes visible; usually found on tree trunks .... LEPIDOPTERA
8b. Body not scaly 9

9a. With a sucking beak; the beak of some may seem to come from between the front legs 10
9b. Beak absent, chewing mouthparts 11

10a. With 2 tube-like projections near the end of the abdomen; soft-bodied and living in colonies on plants; antennae long; beak arises near the front legs .... HEMIPTERA (suborder STERNORRHYNCHA)
10b. Without tube-like projections on abdomen; beak arises from front of head .... HEMIPTERA (suborder HETEROPTERA)

11a. Tarsi either 5-segmented or the hind legs adapted for jumping 12
11b. Tarsi with less than 5 segments and the hind legs not adapted for jumping 14

12a. Hind legs adapted for jumping .... ORTHOPTERA
12b. Hind legs not adapted for jumping 13

13a. Body flattened from top-to-bottom, head hidden from above by thorax .... BLATTODEA
13b. Body stick-like, not flattened; head not hidden by thorax .... PHASMATODEA

14 a. Ant-like appearance, except with soft, white bodies; 4-segmented tarsi; eyeless; antennae resemble a string of round beads; thorax and abdomen are broadly joined .... ISOPTERA
14b. Not fitting the description of 15a; eyes usually well-developed 15

15a. With a forked tail near the end of the body used for jumping; tail may be folded under the body ....COLLEMBOLA
15b. Without a forked tail 16

16a. Oval-shaped and louse-like in appearance; antennae long, thread-like .... PSOCOPTERA
16b. Body narrow; found on leaves and flowers .... THYSANOPTERA

17a. One pair of wings, hind pair reduced to small structures that resemble golf tees ....DIPTERA
17b. With two pairs of wings, although the first pair may be hardened and do not function in flight 18

18a. Front wings thicker in texture than hind wings for all or part of their area 19
18b. Front and hind wings both of the same texture throughout 24

19a. Front wings hard or leathery in texture throughout and almost always meeting in a straight line down the center of the back 20
19b. Front wings parchment-like or leathery throughout or on the basal half only, they do not meet in a straight line down center of back. In lace bugs, the entire top of the insect resembles lace 21

20a. Front wings short, leaving much of the abdomen exposed; a pair of pincher-like appendages extend from the end of the abdomen .... DERMAPTERA
20b. Front wings usually cover all of the abdomen; never with abdominal appendages……… COLEOPTERA

21a. With a jointed beak; basal part of the wing thickened and the tip membranous. Antennae with 5 or less segments .... HEMIPTERA
21b. Chewing mouthparts; front wings parchment-like throughout; antennae with many segments 22

22a. Hind legs adapted for jumping ....ORTHOPTERA
22b. Hind legs not adapted for jumping 23

23a. Front legs adapted for capturing prey (praying mantises) .... MANTODEA
23b. Front legs not adapted for prey; body flattened from top-to-bottom; head hidden from above by thorax .... BLATTARIA

24a. Wings with scales on all or part of their area; mouthparts in the form of a coiled "tongue" ……LEPIDOPTERA
24b. Wings without scales, although they may have hairs 25

25a. Wings long, narrow, veinless, and all 4 are of equal size and have fringes with long hairs; small insects about 1/10 inch long; tarsi 1- or 2-segmented .... THYSANOPTERA
25b. Not fitting the description in 25a 26

26a. Mouthparts composed of a beak arising far back on the underside of the head near the front legs; wings held roof-like over the body, the hind pair smaller than the front pair .... HOMOPTERA
26b. Mouthparts not in the form of a piercing beak, although the front of the head may be prolonged into a longsnout; wings not held roof-like over the body; usually the hind pair of wings are about the same size as the front pair OR the abdomen has 2 or 3 long, thread-like tails 27

27a. With many cross-veins (more than 15) in each wing 28
27b. With few cross-veins, or the veins are indistinct 32

28a. Antennae about as long as the head and thorax together, or longer 30
28b. Antennae short and bristle-like, about as long as head alone or shorter 29

29a. Hind wings much smaller than front wing; occasionally, hind wings absent; abdomen ending in 2 or 3 long, thread-like tails .... EPHEMEROPTERA
29b. Front and hind wings nearly equal in size; no abdominal tails .... ODONATA

30a. Abdomen ending with 2 short tails .... PLECOPTERA
30b. Abdomen without tails 31

31a. Head prolonged into a snout; the tip of the abdomen sometimes resembles a scorpion tail .... MECOPTERA
31b. Head not prolonged into a snout .... NEUROPTERA

32a. All four wings long, narrow, equal-sized, without distinct veins; wings about twice the body length .... ISOPTERA
32b. Not fitting the description in 33a 33

33a. Wings hairy; antennae thread-like and usually as long as or longer than the body; mouthparts indistinct; front and hind wings nearly equal in size .... TRICHOPTERA
33b. Wings not hairy; chewing mouthparts present; hind wings smaller than the front wings 34

34a. Tarsi 2- or 3-segmented; small insects less than 1/8 inch long. Never constricted between the thorax and the abdomen .... PSOCOPTERA
34b. Tarsi 4 or 5 segmented; size variable; most are constricted between the thorax and the abdomen .... HYMENOPTERA